SATHI CHANDRA SEKHAR MANIKANTA REDDY

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STATEMENT OF PURPOSE

Introduction :

I am Sathi Chandra Sekhar Manikanta Reddy, a third-year Computer Science and Engineering undergraduate at Lendi Institute of Engineering and Technology, holding a CGPA of 8.86/10.0. My academic journey is driven by a specific fascination: the intersection of Scalable Distributed Systems and Predictive Artificial Intelligence. I am writing to apply for the Summer Fellowship at IIT Madras to translate my experience in architecting high-throughput data pipelines and deep learning models into rigorous academic research.

Academic Background & Technical Foundation :

My coursework has provided a robust foundation in Data Structures & Algorithms, Machine Learning, Operating Systems, and AWS Cloud Foundations. Beyond the classroom, I have actively validated my skills through industry-recognized certifications, including Google Certified Generative AI and AWS Certified Cloud Practitioner. My proficiency extends to a diverse technology stack, including Python, C++, TensorFlow, PyTorch, and the MERN stack, allowing me to approach problems from both a theoretical and engineering perspective.

Research & Project Experience :

My research potential is best demonstrated through my work in developing intelligent systems that solve real-world efficiency problems:

- Deep Learning & Time-Series Analysis: I architected a Cryptocurrency Price Prediction system using Long Short-Term Memory (LSTM) networks and TensorFlow. By engineering a high-throughput pipeline to ingest and normalize historical data from multiple market APIs, I achieved 80% prediction accuracy on test data, outperforming baseline models. This project honed my skills in handling high-volatility datasets and optimizing model reliability.
- Computer Vision & Edge Computing: To address administrative inefficiencies, I developed an Automated Attendance System using YOLOv8 and FaceNet. I optimized the frame processing with OpenCV to maintain 30+ FPS on edge devices, achieving 99% accuracy in student identification and cutting manual processing time by 90% via an integrated SQLite logging system.
- Distributed Systems & Scalability: I designed a Ride-Sharing Application (Uber Clone) and a Real-Time Bus Tracking System. These projects required engineering scalable backends capable of supporting thousands of concurrent users. I utilized WebSockets for low-latency bi-directional communication and implemented . complex matching algorithms to reduce user wait times, demonstrating my grasp of fault tolerance and data consistency in distributed environments.

Leadership & Algorithmic Excellence :

Research requires not just technical skill, but the ability to communicate complex ideas. As the GeeksForGeeks Campus Mantri. Furthermore, I have demonstrated elite problem-solving abilities by architecting solutions for complex algorithmic challenges on platforms like LeetCode, CodeForces, and GeeksForGeeks.

Proposed Research Plan at IIT Madras During the fellowship, I aim to focus on "Optimizing Transformer Architectures for Real-Time Time-Series Forecasting," leveraging my background with LSTMs and distributed data pipelines. I am eager to explore how attention mechanisms can be optimized for lower latency without sacrificing accuracy in high-frequency data environments. IIT Madras, with its distinguished faculty and research centers like the RBCDSAI, is the ideal environment for me to rigorously formalize these inquiries.

Conclusion : I bring a unique blend of research capability in AI/ML and systems engineering in Cloud/DevOps. I am fully prepared to contribute meaningfully to your lab's ongoing projects and am eager to dedicate my summer to pushing the boundaries of what intelligent systems can achieve.